THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 11, 2023 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (8 points) For any nonempty set A, and for any relations R_1 and R_2 on A, define the *composition* $R_2 \circ R_1$ to be the relation on A given by

$$R_2 \circ R_1 := \{(a, c) \in A \times A : (a, b) \in R_1 \text{ and } (b, c) \in R_2 \text{ for some } b \in A\} \subseteq A \times A.$$

Prove or disprove: If R_1 and R_2 are symmetric, then so is $R_2 \circ R_1$.

The following solution is generated by an AI-based tool (with slight modification of wordings).

If you think you should prove the given statement, comment and find out the missing gap(s) in the following proof (if any).

If you think you should disprove it, point out the mistake(s) and explain briefly what goes wrong. Then provide a disproof.

> "Proof": To show that $R_2 \circ R_1$ is symmetric, we need to show that for any a, c in A, if (a, c) is in $R_2 \circ R_1$, then (c, a) is also in $R_2 \circ R_1$.
> Suppose $(a, c) \in R_2 \circ R_1$. Then there exists $b \in A$ such that $(a, b) \in R_1$ and $(b, c) \in R_2$. Since R_1 is symmetric, we know that (b, a) is also in R_1. Similarly, since R_2 is symmetric, we know that (c, b) is also in R_2. Therefore, we have (b, a) is in R_1 and (c, b) is in R_2, so (c, a) is in $R_2 \circ R_1$. Therefore, $R_2 \circ R_1$ is symmetric.

2. (32 points) Prove or disprove each of the following statements:

   (a) The only prime p in which $p^2 + 8$ is also a prime is $p = 3$.

   (b) The nth root $\sqrt[n]{p}$ of p is irrational for any prime p and any integer $n \geq 2$.

   (c) For any uncountable sets A, B, their intersection $A \cap B$ is not denumerable.

   (d) There exists a nonempty set A and a function $f : A \longrightarrow A$ such that f has no left inverse and no right inverse.

   (e) There exists $n \in \mathbb{N}$ such that the integer $2^n + n^2$ has 5 as its last digit.

3. (18 points)

(a) Consider the sequence $F_1, F_2, F_3, \ldots$ of *Fibonacci numbers* satisfying $F_1 = 1$, $F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for any integer $n \geq 3$. Make a guess as to the value of

$$F_1 + F_3 + \cdots + F_{2n-1}$$

for any $n \in \mathbb{N}$ and prove by induction that your guess is correct.

(b) Let $a_1, a_2, a_3, \ldots$ be a sequence of real numbers satisfying $a_1 = 1$ and

$$a_{n+1} = a_n + 2a_{n-1} + 3a_{n-2} + \cdots + na_1 + 1$$

for any $n \in \mathbb{N}$. Compute the next 4 terms in the sequence, that is, compute a_2, a_3, a_4, a_5.

(c) Make a guess as to the value of a_n for any $n \in \mathbb{N}$ and prove by induction that your guess is correct. (*Hint*: Use the result of (a).)

4. (20 points) Let A be the set of odd integers and B the set of even integers. Define a function $f : A \times A \longrightarrow B \times A$ by

$$f(x, y) = (x - y,\ 2x + y).$$

(a) Prove that f is injective.

(b) Prove that f is not surjective.

(c) Find the range of f and prove your answer.

5. (12 points) Let A be a nonempty finite set and let $f : A \longrightarrow A$ be a function. Suppose that $f \circ f = id_A$, the identity function on A.

(a) Let R be a relation on A defined by

$$aRb \text{ if } a = b \text{ or } f(a) = b.$$

Prove that R is an equivalence relation on A. Find also the equivalence class $[a]$ for any $a \in A$.

(b) Denote by A_f the set of fixed points of f, that is,

$$A_f = \{a \in A : f(a) = a\}.$$

Use (a), or otherwise, prove that $|A|$ and $|A_f|$ must have the same parity.

6. (10 points) Prove that for any nonempty set S, if S is denumerable and $\mathbb{R} \cap S = \emptyset$, then $\mathbb{R}$ is numerically equivalent to $\mathbb{R} \cup S$.
(*Hint*: Construct a bijective function from $\mathbb{R}$ to $\mathbb{R} \cup S$ and prove that the function you constructed is indeed bijective.)

******* **END OF PAPER** *******